

Growth expectations in the AI era

Levels, uncertainty and upper tails in professional GDP forecasts

Max Ghenis

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Abstract

Professional growth forecasts provide a record against which to assess expectations of AI-driven economic acceleration. We examine the joint evolution of forecast levels, stated uncertainty and high-growth probabilities in the US and ECB Surveys of Professional Forecasters. The data extend through 2026Q3; our main historical comparisons hold the survey quarter and calendar horizon fixed. Between 2015–19 and 2025–26, mean next-year US GDP forecasts fell from 2.14% to 1.84%, while reconstructed pooled standard deviation rose from 1.27 to 1.43 percentage points. Reported probability of growth above 4% fell from 5.62% to 4.22%. Similar central forecasts coexist with a qualification in the euro area: its four-year-ahead probability above 4% increased from 0.43% to 1.06%. The reported range probabilities and reconstructed distributions show limited movement toward very rapid growth, rather than universal stability. This is evidence about the overall growth outlook within the surveyed horizons, not an estimate of AI's effect or proof that forecasters ignore AI. We bound extreme-growth probabilities directly from reported bins, retain the COVID episode, and distinguish this evidence from longer-horizon and conditional AI scenarios.

1 The question: where would a growth transformation appear?

A forecast median near 2% need not imply confidence that growth will stay near 2%. Someone who assigns a small but consequential probability to a large acceleration could report an ordinary median alongside an unusual upper tail. Assessing divergence in beliefs about AI and growth therefore requires more than comparing central estimates. It requires looking at distributions and how they have changed.

The Surveys of Professional Forecasters offer that evidence. Their respondents report probabilities over ranges of GDP growth. These distributions allow us to ask three related questions. Have central forecasts moved toward faster growth? Have the distributions widened? Has more probability moved into high-growth outcomes? Our contribution is a reproducible answer across US and euro-area horizons, with a twenty-year historical perspective and an explicit treatment of what the bins can identify.

The dominant pattern is the absence of a large upward shift. Central forecasts are generally below their pre-COVID levels. Uncertainty rose around COVID and subsequently declined, although some measures remain above their earlier levels. The high-growth tail is small in aggregate, with an increase in the euro area that deserves attention. This combination provides more evidence about the expected growth outlook than the narrow dispersion of long-run point forecasts alone.

The interpretation is conditional on the claim being compared. Forecasts that place substantial unconditional probability on rapid US or euro-area growth by the late 2020s or early 2030s face a measurable contrast with these distributions. Scenarios in which advanced AI eventually produces explosive global growth ask a different question. We separate those comparisons rather than place heterogeneous projections on a single numerical scale.

2 Data and comparison design

The underlying pipeline produces 3,695 density summaries across all available variables and 3,070 forecast–outcome pairs. This paper focuses on real GDP growth. US real GDP density forecasts begin in 1992; earlier nominal and real GNP observations remain available in the tracker but do not enter these growth comparisons. ECB GDP densities begin in 1999 (Federal Reserve Bank of Philadelphia 2026; Allayioti et al. 2024).

Our main series are the US next-calendar-year and three-calendar-years-ahead distributions, and the ECB longer-term distribution. We select Q1 rounds throughout. This keeps the calendar horizon fixed and makes the ECB longer-term series a forecast four years ahead. In Q3 and Q4, that question instead refers to a year five years ahead. These are annual growth rates in the target year, not growth averaged over the intervening years.

The main period comparison is 2015–19 versus 2025–26: five pre-COVID Q1 observations and two recent Q1 observations. We show the full trajectory from 2006 where available; US year +3 starts later. The figures retain 2020–24. That window includes both pandemic disruption and the US wide-bin regime from 2020Q2 through 2024Q1. It is a descriptive separation, not a claim that all deviations in those years were caused by COVID.

The companion table also reports 2010–19, 2010–14, 2020–24 and 2023–26 windows and an all-round sensitivity. For that sensitivity, we average within survey year before averaging across years. Thus the three available rounds in 2026 do not automatically give that year less weight than a complete year. All-round comparisons nevertheless mix within-year horizon lengths. The panel can change between rounds; these comparisons do not identify within-person updating or a causal effect of the arrival of generative AI. We do not perform an equivalence test or interpret a short recent sample as proof of an unchanged process.

3 Central forecasts and uncertainty

Table 1 compares means and spreads at fixed horizons. Standard deviation refers to the pooled predictive distribution; the interquartile range is the distance between its 25th and 75th percentiles. Neither is the dispersion of point forecasts across people.

Table 1: Q1 forecast levels and spreads, 2015–19 versus 2025–26. Each cell shows earlier → recent; means are percentages and spreads are percentage points.

Forecast	Mean	Pooled SD	Pooled IQR
US, next year	2.14 → 1.84	1.27 → 1.43	1.41 → 1.60
US, year +3	1.97 → 1.86	1.41 → 1.47	1.64 → 1.53
Euro area, year +4	1.59 → 1.30	0.83 → 0.89	0.98 → 0.96

Mean growth forecasts decline in all three comparisons. The reconstructed widths do not show a radically broader range of outcomes under the stated within-bin and tail assumptions. The US next-year standard deviation rises moderately, while the three-year-ahead and ECB four-year-ahead standard deviations move less. The interquartile ranges at the two longer horizons are slightly narrower. These facts support limited overall movement toward faster and more uncertain growth; they do not support saying every distributional feature is unchanged.

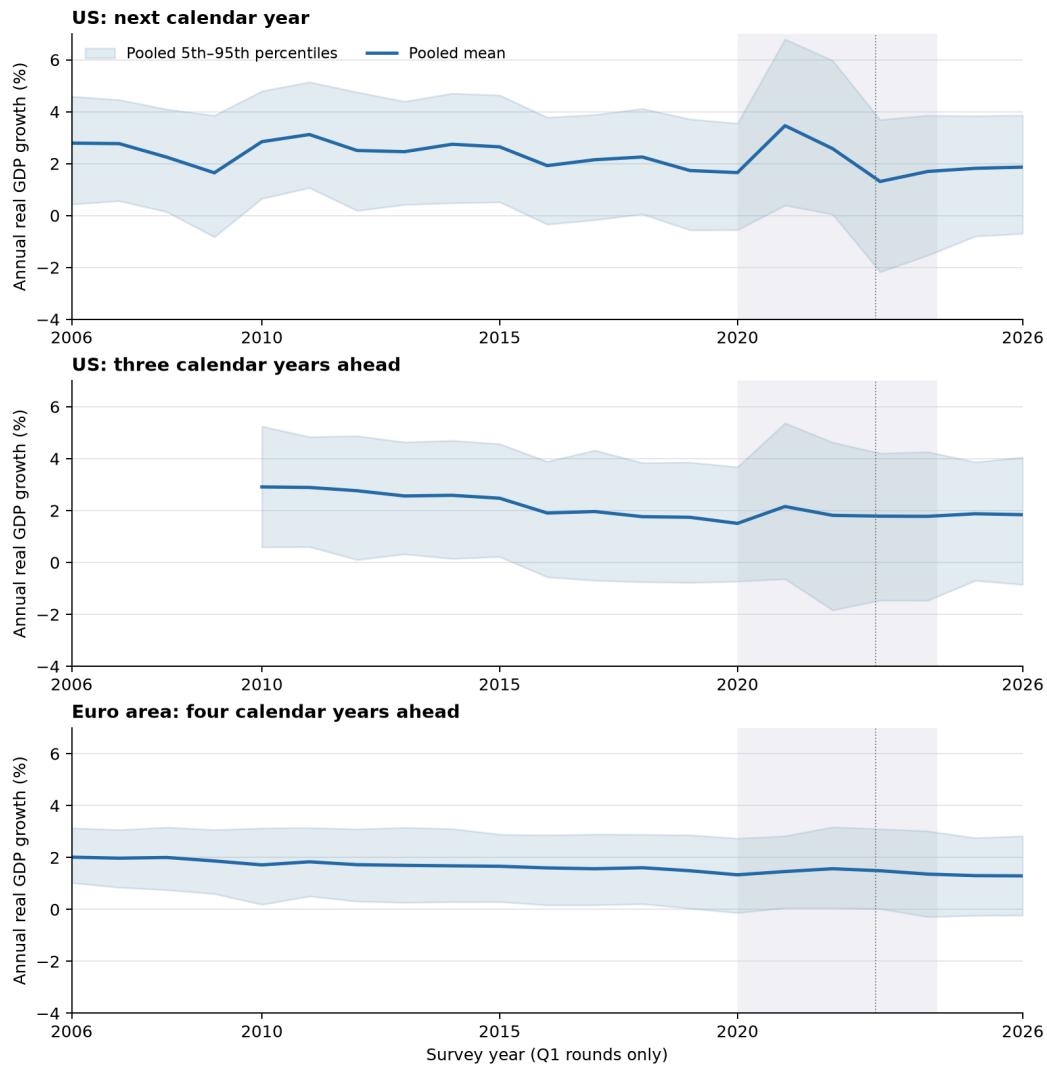


Figure 1: GDP forecast means and reconstructed pooled 90% intervals, Q1 rounds. Gray shading marks 2020–24; the dotted line marks late 2022 as a visual reference, not a treatment date. US year +3 is unavailable before its introduction.

The longer historical record also cautions against describing central forecasts as literally constant over twenty years. The US ten-year median point forecast fell from 3.2% in 2006 to 2.1% in 2026. That forecast concerns average annual growth over a decade. Its cross-sectional spread measures disagreement about a point estimate, not predictive uncertainty over ten years. We therefore report its level separately and make no claim of an inverted uncertainty term structure from comparing it with a one-year distribution.

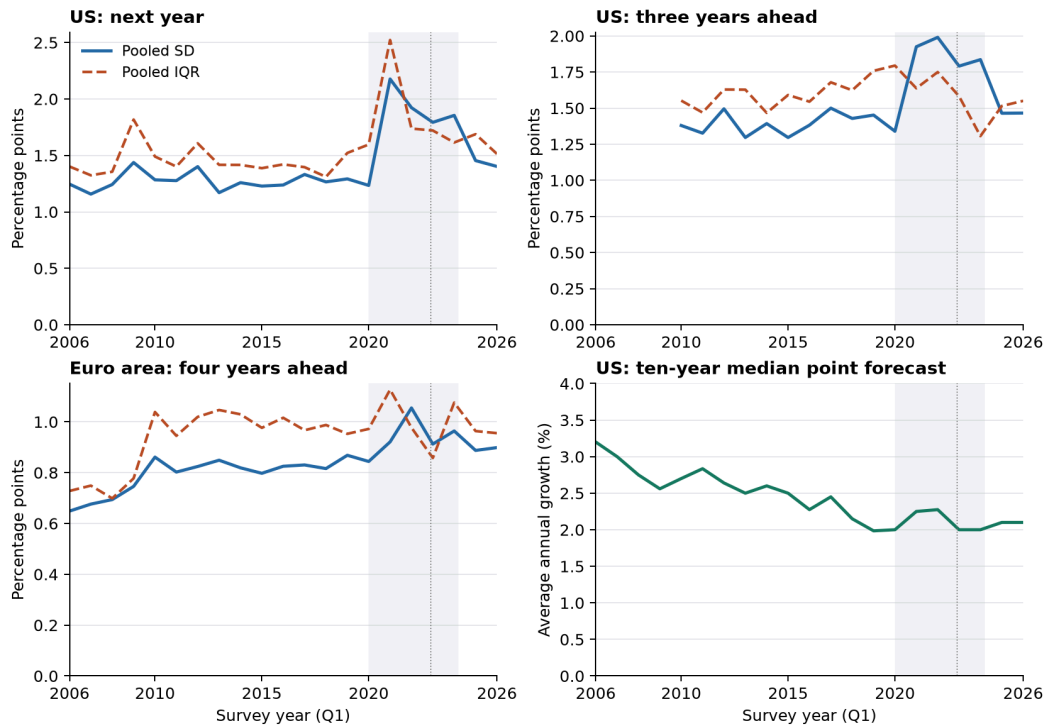


Figure 2: Predictive SD and IQR at three fixed horizons, alongside the US ten-year median point forecast. The final panel is a different outcome and contains no elicited predictive interval.

4 The upper tail: small probabilities, with a euro-area increase

We compute probabilities above 3%, 4%, 5% and 10%. These thresholds are transparent descriptions of outcomes, not definitions of transformative AI. For historical comparisons, 4% is especially useful because it is a common literal bin edge in the displayed series. Its exceedance probability does not require a within-bin interpolation assumption.

Table 2: High-growth probabilities from literal bins, Q1 rounds. Ranges are identification bounds, not sampling confidence intervals.

Forecast	Period	Above 4%	Above 5%	Above 10%
US, next year	2015–19	5.62%	1.19%	0.00–0.32%
US, next year	2025–26	4.22%	0.52–4.22%	0.00–0.05%
US, year +3	2015–19	5.97%	1.69%	0.00–0.45%
US, year +3	2025–26	4.85%	0.85–4.85%	0.00–0.07%
Euro area, year +4	2015–19	0.43%	0.00–0.43%	0.00–0.43%
Euro area, year +4	2025–26	1.06%	0.00–1.06%	0.00–1.06%

The US probability above 4% declines at both horizons. The ECB’s four-year-ahead probability rises from 0.43% to 1.06%. That increase is substantial relative to its starting value, while remaining small in absolute probability. A conclusion that the upper tail did not change anywhere would miss it.

The more extreme thresholds illustrate a measurement limit. In the ECB’s recent scheme, all growth at or above 4% shares one open bin. Its entire probability could lie below 5%, above 10%, or somewhere between. We can bound the probability of those events, but cannot distinguish them. The newer US scheme also places 5% inside a finite bin, widening the identified range relative to earlier schemes. An

apparent precision gain from fitting a distribution inside those bins would come from an assumption rather than another survey response.

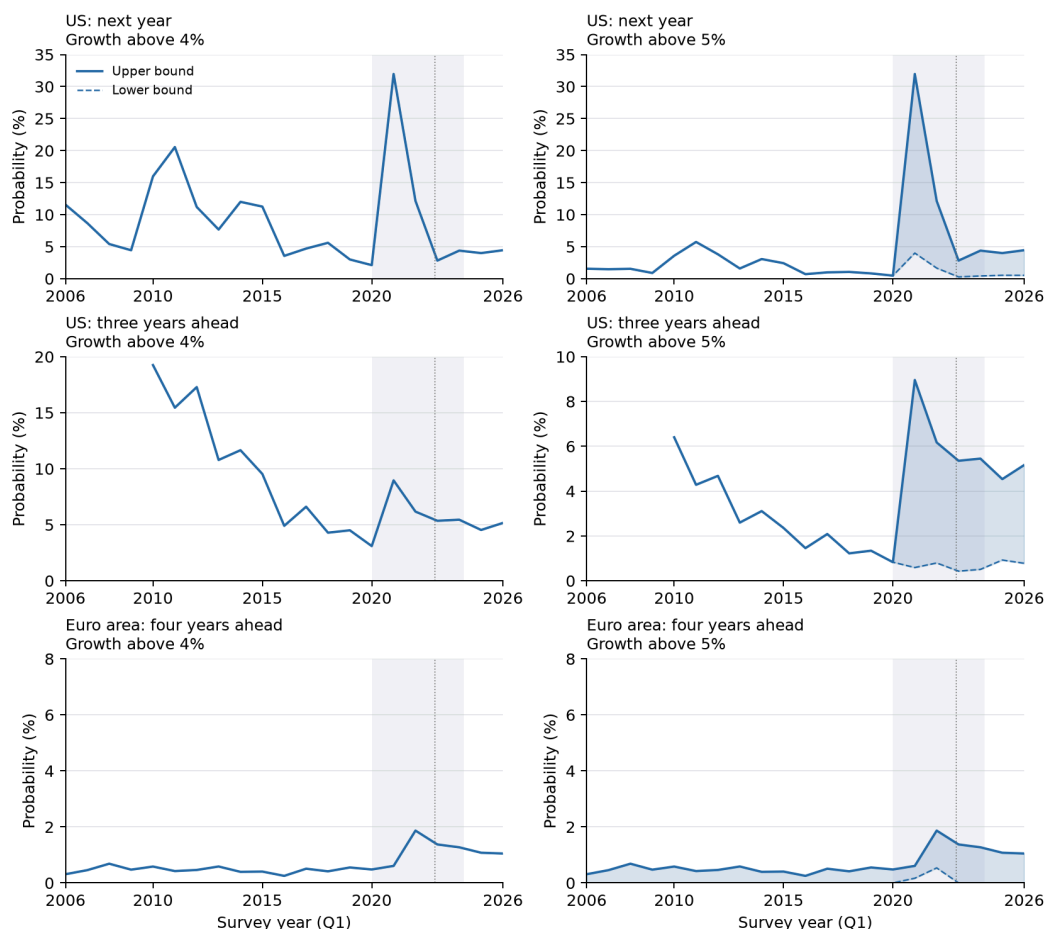


Figure 3: Upper and lower bounds on growth probabilities, Q1 rounds. Where they coincide, the probability is identified at a literal bin edge. The shaded range reflects bin resolution, not uncertainty about the estimated population mean.

The most recent all-round observation is also informative. In 2026Q3, pooled probability above 4% is 3.51% for next-year US growth and 1.45% for ECB longer-term growth, now targeting 2031. In the ECB round, 10 of 27 retained respondents assign positive probability to that bin. A small pooled tail does not mean every respondent rules out high growth.

5 Comparison with AI growth forecasts

The most relevant comparison is with elicited economic distributions, because similar medians can conceal different tails. Karger et al. (2026) elicit US growth beliefs for five-year annualized growth over 2025–29 and 2045–49. Their Figure 5 fits pooled distributions to individual quantiles. For AI experts in 2025–29, the probability above 10% is displayed as 0.0% unconditionally and 3.5% conditional on rapid AI progress. For 2045–49, those figures are 1.4% and 10.4%. These plotted values are rounded. This source does not establish a large unconditional near-term extreme-tail disagreement with SPF. The averaging interval, elicitation and population also differ; a conditional distribution needs scenario probabilities before comparison with unconditional beliefs.

Other prominent estimates differ more fundamentally in outcome or horizon:

Source	Outcome and interpretation	Relation to this paper
Acemoglu (2024)	Task-based estimate of AI's contribution to productivity over ten years	An incremental productivity effect, not a probability distribution for annual GDP growth
Davidson (2021)	Possibility of global growth above 30% per year before 2100	Shows why extreme outcomes matter; lies beyond the SPF geography and horizon
Erdil and Besiroglu (2023)	Arguments for explosive growth from broad AI automation	Connects assumptions about automation to growth regimes; does not supply a matched annual US SPF probability

The contribution of the SPF evidence is specific. The observable professional outlook has not shifted toward substantially faster growth in the center, and its reported upper-tail mass places quantitative limits on high-growth outcomes within its horizons. This would contrast with beliefs assigning appreciable unconditional probability to such outcomes over those same horizons. The present source comparison does not identify a single commensurable probability difference between two populations, and we do not manufacture one by treating global, conditional or century-scale scenarios as US annual forecasts.

There are several ways the outlook could remain broadly similar while respondents revise their views about AI. Positive AI effects could offset demographic or other economic headwinds. Respondents could expect deployment after the forecast horizon. A subset could revise upward while others revise downward. Survey bins and rounding could obscure changes in very small probabilities. These are possible explanations, not mechanisms established by the data. Establishing them requires matched questions and respondent-level evidence on AI assumptions.

6 What the historical calibration adds

Historical evaluation helps assess how much evidential weight to place on the stated probabilities. It does not establish which view of AI is correct. For Q1 next-year US GDP, a band of one pooled standard deviation around the mean covers 23 of 33 outcomes. The pooled 90% interval covers 26 of 33 (78.8%). The latter falls below its nominal coverage. One-standard-deviation coverage is not a general calibration test for an arbitrary non-Gaussian distribution. The misses include major economic disruptions and the late-1990s growth acceleration.

We also evaluate entire distributions with the continuous ranked probability score (CRPS) (Gneiting and Raftery 2007). For each benchmark, skill is one minus mean forecast CRPS divided by mean benchmark CRPS on the same eligible observations. This compares average loss directly. Averaging per-observation percentage skill gives a different weighting of outcomes, so we retain it as a sensitivity rather than interpret its sign as a definitive finding about useful information.

Table 3: GDP CRPS skill against expanding benchmarks, all scored horizons. Positive values favor the survey. Each benchmark uses its own matched eligible sample.

Series	Benchmark	Rows	Ratio of mean CRPS	Mean row skill
US GDP	climatology	382	38.2%	40.4%
US GDP	gaussian	312	6.4%	6.2%
Euro-area GDP	climatology	489	19.6%	15.3%
Euro-area GDP	gaussian	480	0.9%	17.1%

Forecast–outcome pairs can share a realized target, and rolling outcomes overlap. These summaries are descriptive and do not establish statistical significance. The Gaussian benchmark learns the RMSE of prior forecast errors; climatology uses prior outcomes. Both require ten historical observations. Completed target periods must precede the forecast quarter, but actual historical publication dates and vintages are not reconstructed. This is an expanding-window evaluation with archived or newly acquired revised outcomes, not a real-time backtest.

7 Measurement and reproducibility

Within a round, we retain respondent histograms with probabilities between 0 and 100 and totals strictly within two percentage points of 100. Partial missing entries count as zero; entirely missing histograms are excluded. Retained probabilities are normalized and respondents receive equal weight. This rejects two negative ECB entries in 2023Q1 as well as invalid total-probability responses.

For moments, quantiles and scores, probability is uniform inside each finite bin. An open tail receives the width of its nearest finite positive-width bin. Literal gaps between bins receive zero probability. If bin b has midpoint x_b and width w_b , respondent i has

$$\mu_i = \sum_b p_{ib} x_b, \quad v_i = \sum_b p_{ib} \left((x_b - \mu_i)^2 + \frac{w_b^2}{12} \right).$$

The equal-weight empirical mixture has variance

$$V = \frac{1}{n} \sum_i v_i + \frac{1}{n} \sum_i (\mu_i - \bar{\mu})^2.$$

The first component is average individual uncertainty; the second is disagreement (Wallis 2005; Zarnowitz and Lambros 1987). The denominator is n , because the object is the finite empirical pool. Using a sample-variance correction would estimate a different object. Independent numerical-integration tests check the identity against the pooled distribution. The reported within SD is the square root of average individual variance, not average individual SD.

For high-growth probabilities, we dispense with finite-tail closure. If a threshold falls inside a bin, its entire mass contributes to the upper bound and none to the lower bound. Bins entirely above the threshold contribute to both. At exact edges we assume continuous outcomes, so a singleton boundary has no probability mass. Averaging respondent bounds yields pooled bounds. An optional uniform interpolation is emitted only when the threshold is outside both open tails; it is not used for the headline bounds. Small mass in an open tail does not bound the magnitude of growth conditional on reaching that tail. Consequently, the bins alone do not identify an unrestricted mean or variance: the level and width results depend on the reconstruction, while the exceedance bounds do not.

These distinctions matter across survey redesigns. Neither SD nor IQR is fully invariant to changing bins. Common-edge probabilities provide the cleanest historical tail comparison. Very small reported probabilities are also limited by respondent rounding; zero reported mass is evidence about the elicited answer, not proof that the respondent’s latent subjective probability is exactly zero.

Annual US outcomes use the archived BEA series. Annual ECB GDP outcomes use growth in sums of complete quarterly GDP levels, replacing the earlier approximation that averaged quarterly year-over-year rates. Calendar inflation uses the official annual-average index-growth series, published to 0.1 percentage point. Rolling observations retain the archived files. The annual companion data were retrieved on 5 September 2026; original rolling retrieval dates are unrecorded. The GDP levels reproduce the archived quarterly growth values, while inflation’s annual status can differ from archived monthly estimate flags. URLs, hashes, precision and verification checks are recorded in [data/raw/ecb_annual_realizations_sources.json](#).

The repository contains raw snapshots, reusable parsers, independent numerical tests, [growth_tails.csv](#), period comparisons, score summaries, figure-generation code and the Quarto manuscript. Regenerating the analysis requires the pipeline build, the research figure build, the tracker data build and the manuscript render. The local research companion at [site/growth/index.html](#) provides baseline, round and threshold controls; the source repository contains the existing public release.

8 Conclusion

The strongest evidence is the joint behavior of forecast levels, widths and tails. The professional outlook in these surveys remains centered on modest growth, with limited movement toward much faster outcomes at available horizons. That statement survives looking beyond point estimates. It requires a qualification: the ECB’s small longer-term high-growth tail has increased.

This record makes the professional baseline for comparisons of near-term growth beliefs concrete. It cannot establish that respondents have neglected AI or resolve projections beyond the surveyed horizons. A direct comparison would ask professional macroeconomic forecasters and AI specialists for distributions over the same US annual and multi-year growth outcomes, on the same dates, both unconditionally and conditional on stated AI capabilities. The existing surveys provide a measurable baseline for that comparison.

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